Configurable Value and Data for Analytics

Configurable Value:

Token:

1. On boarding token: 11 tokens.
2. Deposit from seeker: 1 token.
3. Successful swap: 2 tokens.
4. Seeker no show: 1 token.
5. Transferable token: no limit provided there is enough token.
6. Lending token: 2 tokens each time, 4 tokens per day. Maximum 10 tokens at all time.
7. Repayment of token: any amount capped at maximum 10 tokens.

Time:

1. Hosting time (leaving in 5 mins): 300 seconds.
2. Hosting time (leaving in 15 mins): 900 seconds.
3. Respond time to reject automatch: 30 seconds.
4. Giver extension of time: 5 mins x 60 seconds: 300 seconds.
5. Response period after end of hosting time for giver to respond: 60 seconds
6. Response period after seeker receives notification that giver has responded (successful swap or seeker no show): 300 seconds.
7. Grace period to repay tokens: 7 days x 24 hours x 60 minutes x 60 seconds: 604,800 seconds
8. Grace period to submit dispute: 24 hours x 60 mins x 60 seconds from end of transaction: 86,400 seconds.
9. Admin team response period: 48 hours from submission of dispute.

Data collection:

To measure and implement the success metrics, need a combination of backend logging, data analytics, and dashboard visualization. Here's how you can track and implement each metric:

Note: monthly calculation approach:

1. Take the **total registered users at 12:00 AM UTC** on the 1st of each month.
2. Compare the count of users on the **1st of the current month** to the **1st of the previous month**.

Example Calculation

- **January 1st user count**: 5,000
- **February 1st user count**: 5,500

1. Growth of Users

Implementation:

- Track **total registered users** over time.
- Store timestamps of each user registration.
- Implement a **monthly growth rate calculation**:
- $$\text{Growth Rate} = \frac{\text{New users in current month} - \text{New users in previous month}}{\text{New users in previous month}} \times 100$$
- Use a dashboard (e.g., Google Data Studio, Power BI, or a custom admin panel) to visualize trends.

2. Average Time Spent on App Per Month

Implementation:

- Log each session's **start time** and **end time**.
- Sum up the duration per user for the past month.
- Aggregate across all users to calculate the average:
- $$\text{Avg Time Spent} = \frac{\sum(\text{session duration per user})}{\text{total active users in 30 days}}$$
- Store and display trends in an analytics dashboard.

3. Total Registered Users

Implementation:

- Track and store the count of users who have completed onboarding (phone verification + profile setup).
- Display **cumulative user count** in the admin panel.

4. Monthly Active Giver Usage

Definition: Measures how many times a unique user submitted a parking host request in the past 30 days.

Implementation:

- Log **each hosting submission** (host parking event).
- Count **unique users** who submitted at least once in the last month.
- Store data and display it in the admin panel.

5. Monthly Active Seeker Usage

Definition: Measures how many times a unique user submitted a parking find request in the past 30 days.

Implementation:

- Log **each parking search request**.
- Count **unique users** who submitted a request at least once in the last month.
- Store and display the trend.

6. Monthly Active Users (Host and Find Overlapping)

Definition: Measures how many times a user either hosted or found parking at least once in the past month.

Implementation:

- Combine logs for **hosting and seeking events**.
- Deduplicate by **unique user ID**.
- Store and visualize trends.

7. Monthly Average Usage Per User

Definition: Measures the average number of times an active user used the app for hosting or seeking.

Implementation:

- Calculate:
- $$\text{Monthly Average Usage} = \frac{\text{Total usage (host + seek) in 30 days}}{\text{Total monthly active users}}$$
- Store and display the trend.

8. New User Signup Rate

Definition: Measures how many new users registered in the past month.

Implementation:

- Store **registration timestamp** per user.
- Filter users who registered within the past month.
- Display in analytics.

9. Inactive Users (30 Days of Inactivity)

Definition: Measures users who were active in the past but have not used the app for the last 30 days.

Implementation:

- Check **last activity timestamp** for each user.
- Count users who had activity **before the last Month** but not in the last month.
- Display trends and set alerts.

Tools & Tech for Implementation

1. **Database Logging**
   - Store all user actions in a **PostgreSQL or Firebase Firestore** database.
   - Implement event-driven logging with **Kafka or Redis** for real-time tracking.
2. **Analytics & Reporting**
   - Use **Google Analytics, Mixpanel, or Amplitude** for event tracking.
   - Generate reports using **Power BI, Looker, or Metabase**.
3. **Dashboard for Admins**
   - Build a simple dashboard using **React + Node.js** to display KPIs.
   - Use **Superset or Metabase** for visualization.
4. **Automation & Alerts**
   - Set **email alerts** for abnormal trends (drop in active users, spikes in inactivity).
   - Implement **cron jobs** to update metrics daily.